

Chapter 15

Limits and Derivatives

Only One Option Correct Type Questions

1. $\frac{d^2x}{dy^2}$ equals [IIT-JEE-2007 (Paper-2)]
- (A) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (B) $-\left(\frac{d^2y}{dx^2}\right)^{-1} \left(\frac{dy}{dx}\right)^{-3}$
- (C) $\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-2}$ (D) $-\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$
2. Let $g(x) = \log f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = x f(x)$. Then, for $N = 1, 2, 3, \dots$, $g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$ [IIT-JEE-2008 (Paper-2)]
- (A) $-4\left\{1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2}\right\}$ (B) $4\left\{1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2}\right\}$
- (C) $-4\left\{1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2}\right\}$ (D) $4\left\{1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2}\right\}$
3. The value of $\lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x \frac{t \ln(1+t)}{t^4 + 4} dt$ is [IIT-JEE-2010 (Paper-1)]
- (A) 0 (B) $\frac{1}{12}$
- (C) $\frac{1}{24}$ (D) $\frac{1}{64}$
4. Let f be a real-valued function defined on the interval $(-1, 1)$ such that $e^{-x}f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$, for all $x \in (-1, 1)$, and let f^{-1} be the inverse function of f . Then $(f^{-1})'(2)$ is equal to [IIT-JEE-2010 (Paper-2)]
- (A) 1 (B) $\frac{1}{3}$
- (C) $\frac{1}{2}$ (D) $\frac{1}{e}$
5. If $\lim_{x \rightarrow 0} [1 + x \ln(1 + b^2)]^{1/x} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is [IIT-JEE-2011 (Paper-2)]
- (A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$
- (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$

6. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then

[IIT-JEE-2012 (Paper-1)]

(A) $a = 1, b = 4$

(B) $a = 1, b = -4$

(C) $a = 2, b = -3$

(D) $a = 2, b = 3$

7. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation

$$(\sqrt[3]{1+a}-1)x^2 + (\sqrt{1+a}-1)x + (\sqrt[6]{1+a}-1) = 0$$

where $a > -1$. Then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are

[IIT-JEE-2012 (Paper-2)]

(A) $-\frac{5}{2}$ and 1

(B) $-\frac{1}{2}$ and -1

(C) $-\frac{7}{2}$ and 2

(D) $-\frac{9}{2}$ and 3

One or More Option(s) Correct Type Questions

8. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}, a > 0$

If L is finite, then

[IIT-JEE-2009 (Paper-1)]

(A) $a = 2$

(B) $a = 1$

(C) $L = \frac{1}{64}$

(D) $L = \frac{1}{32}$

9. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$, $\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$.

Then $a =$

[JEE (Adv)-2013 (Paper-2)]

(A) 5

(B) 7

(C) $-\frac{15}{2}$

(D) $-\frac{17}{2}$

10. Let $f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$ for $x \neq 1$. Then

[JEE (Adv)-2017 (Paper-2)]

(A) $\lim_{x \rightarrow 1^+} f(x) = 0$

(B) $\lim_{x \rightarrow 1^-} f(x) = 0$

(C) $\lim_{x \rightarrow 1^+} f(x)$ does not exist

(D) $\lim_{x \rightarrow 1^-} f(x)$ does not exist

11. Let $f : (0, \pi) \rightarrow \mathbb{R}$ be a twice differentiable function such that $\lim_{t \rightarrow x} \frac{f(x) \sin t - f(t) \sin x}{t - x} = \sin^2 x$ for

$x \in (0, \pi)$. If $f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}$, then which of the following statement(s) is (are) TRUE? [JEE (Adv)-2018 (Paper-2)]

(A) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$

(B) $f(x) < \frac{x^4}{6} - x^2$ for all $x \in (0, \pi)$

(C) There exists $\alpha \in (0, \pi)$ such that $f'(\alpha) = 0$

(D) $f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$

12. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be functions satisfying

$$f(x+y) = f(x) + f(y) + f(x)f(y) \text{ and } f(x) = xg(x)$$

for all $x, y \in \mathbb{R}$. If $\lim_{x \rightarrow 0} g(x) = 1$, then which of the following statements is/are TRUE?

[JEE (Adv)-2020 (Paper-2)]

(A) f is differentiable at every $x \in \mathbb{R}$

(B) If $g(0) = 1$, then g is differentiable at every $x \in \mathbb{R}$

(C) The derivative $f'(1)$ is equal to 1

(D) The derivative $f'(0)$ is equal to 1

Integer / Numerical Value Type Questions

13. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is

[IIT-JEE-2009 (Paper-2)]

14. Let $p(x)$ be a polynomial of degree 4 having extremum at $x = 1, 2$ and $\lim_{x \rightarrow 0} \left(1 + \frac{p(x)}{x^2}\right) = 2$. Then the value of $p(2)$ is

[IIT-JEE-2009 (Paper-2)]

15. The largest value of the non-negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is

[JEE (Adv)-2014 (Paper-1)]

16. Let m and n be two positive integers greater than 1. If $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n)} - e}{\alpha^m} \right) = -\left(\frac{e}{2}\right)$, then the value of $\frac{m}{n}$ is

[JEE (Adv)-2015 (Paper-2)]

17. Let $\alpha, \beta \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals

[JEE (Adv)-2016 (Paper-1)]

18. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f'(0) = 1$. If

$$g(x) = \int_x^{\frac{\pi}{2}} [f'(t) \operatorname{cosec} t - \cot t \operatorname{cosec} t f(t)] dt \text{ for } x \in \left(0, \frac{\pi}{2}\right), \text{ then } \lim_{x \rightarrow 0} g(x) =$$

[JEE (Adv)-2017 (Paper-1)]

19. For each positive integer n , let $y_n = \frac{1}{n}((n+1)(n+2)\dots(n+n))^{\frac{1}{n}}$. For $x \in \mathbb{R}$, let $[x]$ be the greatest integer less than or equal to x . If $\lim_{n \rightarrow \infty} y_n = L$, then the value of $[L]$ is ____.

[JEE (Adv)-2018 (Paper-1)]

20. The value of the limit

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x + \sin x)}{\left(2 \sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}\right) - \left(\sqrt{2} + \sqrt{2} \cos 2x + \cos \frac{3x}{2}\right)} \text{ is } \underline{\hspace{2cm}}$$

[JEE (Adv)-2020 (Paper-2)]

21. If $\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1-x^3)^{\frac{1}{3}} + ((1-x^2)^{\frac{1}{2}} - 1) \sin x}{x \sin^2 x}$, then the value of 6β is ____.

[JEE (Adv)-2022 (Paper-2)]

